Ch 4.4

1.

Orthogonal Vectors: Two vectors are orthogonal if their dot product is zero.

Orthonormal Vectors: Two vectors are orthonormal if they are orthogonal and each of them has a unit length (magnitude of 1).

Independent Vectors: Two vectors are independent if they are not scalar multiples of each othe

a) $\begin{bmatrix}1\\0\end{bmatrix} \cdot \begin{bmatrix}-1\\1\end{bmatrix} = -1 - 0 \neq 0$ to be O. Norm

$\begin{bmatrix}1\\0\end{bmatrix}$ $\begin{bmatrix}0\\1\end{bmatrix}$

indep.

$.24 \div 0.24 = 0$

b) $\begin{bmatrix}.6\\.8\end{bmatrix} \cdot \begin{bmatrix}4\\-.3\end{bmatrix}$

$\rightarrow \begin{bmatrix}.6\\.8\end{bmatrix}$ $\begin{bmatrix}.8\\-.6\end{bmatrix}$

orthogon + indep

$\sqrt{0.36 + 0.64}$

$= 1.$

c) $\begin{bmatrix}\cos\theta\\\sin\theta\end{bmatrix}$ $\begin{bmatrix}-\sin\theta\\\cos\theta\end{bmatrix}$

O.N. + indep !

② $q_1 = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)$, $q_2 = \left(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$?

$\begin{bmatrix}0\\0\end{bmatrix}$

$Q^TQ = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$QQ^T = \begin{bmatrix} 5/9 & 2/9 & -4/9 \\ 2/9 & 8/9 & 2/9 \\ -4/9 & 2/9 & 5/9 \end{bmatrix}$

(3) a) 3 Orthogonal col → each len 4.

$A^TA = 16I$

b) 3 " " → 1, 2, 3 len

A^TA will diag entries $1^2, 2^2, 3^2$

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Part (a): If A has three orthogonal columns each of length 4, what is A^TA?

1. Orthogonality and length of columns:
   - The columns of A are orthogonal, meaning their dot product with each other is zero. So, if $A = [c_1, c_2, c_3]$, where c_1, c_2, c_3 are the columns, we have:

$$c_1^T c_2 = 0, \quad c_1^T c_3 = 0, \quad c_2^T c_3 = 0$$

1. Length of each column:
   - The length of each column is 4. Hence, the dot product of each column with itself is 16:

$$c_1^T c_1 = 16, \quad c_2^T c_2 = 16, \quad c_3^T c_3 = 16$$

1. Form of A^TA:
   - Since A^TA is the matrix of dot products between the columns of A, we know:

$$A^TA = \begin{bmatrix} c_1^T c_1 & c_1^T c_2 & c_1^T c_3 \\ c_2^T c_1 & c_2^T c_2 & c_2^T c_3 \\ c_3^T c_1 & c_3^T c_2 & c_3^T c_3 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 16 \end{bmatrix}$$

Part (b): If A has three orthogonal columns of lengths 1, 2, and 3, what is A^TA ?

1. Orthogonality and length of columns:
   - The columns of A are orthogonal, meaning their dot product with each other is zero. So, the dot products are:

$$c_1^Tc_2 = 0, \quad c_1^Tc_3 = 0, \quad c_2^Tc_3 = 0$$

1. Length of each column:
   - The length of each column is given as 1, 2, and 3. Hence, the dot product of each column with itself is:

$$c_1^Tc_1 = 1^2 = 1, \quad c_2^Tc_2 = 2^2 = 4, \quad c_3^Tc_3 = 3^2 = 9$$

Ask anything

④ Q0. Norm Q. But $QQ^T \neq I$

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}_{3\times 2}, \quad QQ^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq I$$

Any Q with $m > n$. $QQ^T \neq I$

P.S: if $m < n$ rank $= m$

If **n > m** (i.e., the matrix **Q** has more rows than columns), the matrix **QQ^T** would still not equal the identity matrix **I**. In this case, the resulting matrix **QQ^T** would generally be a projection matrix as well, but the rank of **QQ^T** would be at most **m**, reflecting the fact that the transformation is not onto the full space **R^n**.

For example, if **Q** is a **3 × 2** matrix, **QQ^T** will be a **3 × 3** matrix, but it will not be an identity matrix. Instead, it will have a rank of 2 (since **Q** has 2 columns), and it will project onto a 2-dimensional subspace of **R^3**.

4 b) (1,0) & (0,0) orthogon + dependent

c) $\frac{1}{\sqrt{3}}(1,1,1)$, $\frac{1}{\sqrt{2}}(1,-1,0)$, $\frac{1}{\sqrt{6}}(1,1,-2)$

(5) plane → $x+y+2z=0$
normal vect → $(1,1,2)$

$x = -y-2z$

putting $y=1, z=0$

$x = -1$

$\therefore V_1(-1,1,0)$ NormF: $\frac{1}{\sqrt{2}}(-1,1,0$

Now $V_2(a,b,c)$

$(1,1,2)(a,b,c)$

$v_1 \cdot v_2 \; (-1, 1, 0) \cdot (\quad)$

$a = b$

as v_2 spans the plane

$a + b + 2c = a + a + 2c = 0$

$\therefore \; c = -a$

$v_2 \; (1, 1, -1)$

normal form: $\frac{1}{\sqrt{3}} (1, 1, -1)$

(6) $(Q_1 Q_2)^T (Q_1 Q_2) = Q_1^T Q_2^T \; Q_1 Q_2$

$= I.$

(7) $Q \rightarrow$ o Norm cols.

L.Sq. Soln $\hat{x}$ to $Qx = b \Rightarrow$

$\therefore \; Q^T Q \hat{x} = Q^T b$

$\therefore \; \hat{x} = Q^T b$

✅ No matrix inversion, no solving a big system — just compute **Q^T b**, and you're done.

✅ "No cost" in solving the normal equations — because you've already paid the price by orthonormalizing with Gram-Schmidt!

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Given matrix A and vector $\mathbf{b}$, solve the least squares problem:

$$A\mathbf{x} \approx \mathbf{b}$$

We'll do this via Gram-Schmidt orthonormalization of A to get Q, then solve using:

$$\mathbf{x}_b = Q^\top \mathbf{b}$$

Step 1: Let's Define

Let's choose a simple example:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Step 2: Apply Gram-Schmidt to Columns of A

We want to find orthonormal vectors $\mathbf{q}_1, \mathbf{q}_2$ to form matrix Q.

1️⃣ First column $\mathbf{a}_1 = [1\ 1\ 0]$

Normalize:

$$\mathbf{q}_1 = \frac{\mathbf{a}_1}{\|\mathbf{a}_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

2️⃣ Second column $\mathbf{a}_2 = [1\ 0\ 1]$

Project $\mathbf{a}_2$ onto $\mathbf{q}_1$:

$$\text{proj}_{\mathbf{q}_1}(\mathbf{a}_2) = (\mathbf{q}_1^\top \mathbf{a}_2)\mathbf{q}_1 = \left(\frac{1}{\sqrt{2}} \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 + 0 \cdot 1\right) \cdot \mathbf{q}_1 = \frac{1}{\sqrt{2}} \cdot \mathbf{q}_1$$

Subtract projection:

$$\mathbf{u}_2 = \mathbf{a}_2 - \text{proj}_{\mathbf{q}_1}(\mathbf{a}_2) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \\ 1 \end{bmatrix}$$

Ask anything

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Subtract projection:

$$\mathbf{u}_2 = \mathbf{a}_2 - \text{proj}_{\mathbf{q}_1}(\mathbf{a}_2) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \\ 1 \end{bmatrix}$$

Normalize:

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|} = \frac{1}{\sqrt{1.5}} \begin{bmatrix} 0.5 \\ -0.5 \\ 1 \end{bmatrix}$$

Step 3: Construct Orthonormal Matrix Q

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{0.5}{\sqrt{1.5}} \\ \frac{1}{\sqrt{2}} & \frac{-0.5}{\sqrt{1.5}} \\ 0 & \frac{1}{\sqrt{1.5}} \end{bmatrix}$$

Step 4: Compute $\mathbf{x}_b = Q^\top \mathbf{b}$

Let's do this:

$$Q^\top \mathbf{b} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{0.5}{\sqrt{1.5}} & \frac{-0.5}{\sqrt{1.5}} & \frac{1}{\sqrt{1.5}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Break it down:

- First entry: $\frac{1}{\sqrt{2}}(1+1) = \frac{2}{\sqrt{2}} = \sqrt{2}$
- Second entry: $\frac{0.5}{\sqrt{1.5}}(1) + \frac{-0.5}{\sqrt{1.5}}(1) + \frac{2}{\sqrt{1.5}} = 0 + 0 + \frac{2}{\sqrt{1.5}}$

So:

$$\mathbf{x}_b = \begin{bmatrix} \sqrt{2} \\ \frac{2}{\sqrt{1.5}} \end{bmatrix}$$

Ask anything

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(8) q_1, q_2 O.Norm vectors in R^5.

q_1 & q_2's comb^n closest to given veet b ⇒

$(q_1^T b)\, q_1 + (q_2^T b)\, q_2$ is " " b

error $e = b - p$ is orthogonal to

q_1 & q_2.

$q_1, q_2 \in \mathbb{R}^5$ are orthonormal, so:

$q_1^T q_1 = 1,$
$q_2^T q_2 = 1,$
$q_1^T q_2 = 0$

Vector $b \in \mathbb{R}^5$

🔶 **Projection onto the span of q_1, q_2:**

$p = (q_1^T b)\, q_1 + (q_2^T b)\, q_2$

This gives you the closest point in the plane spanned by q_1, q_2 to the vector b.

a) $p = QQ^T$; $q_1 = (.8, .6, 0)$ $q_2 = (-.6, .8, 0)$

$$Q = \begin{bmatrix} .8 & -.6 \\ .6 & .8 \\ 0 & 0 \end{bmatrix} \qquad P = QQ^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$p^2 = p$

↳ projection on xy plane.

b) $(QQ^T)^2 = (QQ^T)(QQ^T)$

$= QIQ^T = QQ^T$

(10) O.Norm ... indep.

Proof:

a) Vector Proof: if q_1, q_2, q_3 O.Norm

→ only soln is trivial 0.

then, $c_1q_1 + c_2q_2 + c_3q_3 = 0$

$c_1 = c_2 = c_3 = 0$ → gotta proof

so, $q_1^T(c_1q_1 + c_2q_2 + c_3q_3) = q_1^T 0 = 0$

$c_1q_1^Tq_1 + c_2q_1^Tq_2 + c_3q_1^Tq_3 = 0$

* $q_1^Tq_1 = 1$

$q_1^Tq_2 = 0$

$q_1^Tq_3 = 0$

so, $c_1 \cdot 1 + 0 + 0 = 0$

$\therefore c_1 = 0$

Similarly $c_2 = c_3 = 0$ → None can be written as comb^n of other.

$\therefore q_1, q_2, q_3$ lin. indep.

linear comb^n = 0 only if coeff 0.

b) Matrix Proof: $Q \in R^{n \times m}$

$$Q = [q_1 \; q_2 \; \cdots \; q_m]$$

$$Qx = 0$$

goal for show. $x = 0$

$$Q^T Q x = Q^T \cdot 0$$

$$x = 0$$

No nonzero comb^n of col's give 0.

$\Rightarrow$ linearly ind.

(11) Find O.Norm vects q_1 & q_2 in the plane spanned by,

$a = (1, 3, 4, 5, 7)$, $b = (6, 6, 8, 0, 8)$

$\Rightarrow$ <u>S1: Normalize to get q_1:</u>

$$q_1 = \frac{a}{|a|}$$

$$|a| = \sqrt{1+9+16+25+49} = \sqrt{100} = 10$$

$$q_1 = \frac{1}{10}(1, 3, 4, 5, 7)$$

S2: make b orthogon to q_1 to get q_2

$q_1^T b$

$$\text{proj}_{q_1} b = (b \cdot q_1) q_1$$

$$b \cdot q_1 = -6 \cdot \frac{1}{10} + 6 \cdot \frac{3}{10} + 8 \frac{4}{10} + 0 + 8 \cdot \frac{7}{1}$$

$$= 10$$

$$\text{Now}, (b \cdot q_1) q_1 = 10 \cdot q_1$$

$$= (1, 3, 4, 5, 7)$$

S2.1: Subtract b from $\text{proj}_{q_1} b$: (comp of b orthog to q_1)

$e = b - p$

$$b - \text{proj}_{q_1} b = -7, 3, 4, -5, 1$$

S2.2: Normalize to get q_2

$$|b - \text{Proj}_{q_1} b| = \sqrt{49 + 9 + 16 + 25 + 1} = 10$$

$\therefore q_2 = \frac{1}{10}(-7, 3, 4, -5, 1)$

6) vector in plane closest to $(1,0,0,0,0)$

---

is the proj of v onto subspace spanned by q_1 & q_2

$$\text{proj}_{\text{span}(q_1,q_2)} v = (v \cdot q_1) q_1 + (v \cdot q_2) q_2$$

($a_1^T v$)

$v \cdot q_1 = 1\left(\frac{1}{10}\right) + 0 = \frac{1}{10}$

$v \cdot q_2 = 1\left(-\frac{7}{10}\right) = -\frac{7}{10}$

So, $\frac{1}{10} q_1 - \frac{7}{10} q_2$

$= \frac{1}{10}\left(\frac{1}{10}, \frac{3}{10} - \cdots\right) \quad -\frac{7}{10}\left(-\frac{7}{10}, \frac{3}{10} - \cdots\right)$

$= \left(\frac{1}{100}, \frac{3}{100}, \frac{4}{100}, \frac{5}{100}, \frac{7}{100}\right) +$

$\left(\frac{49}{100}, \frac{-21}{100}, -28, \frac{35}{100}, \right.$

$$\left(\frac{1}{100}, \frac{100}{100}, \frac{}{100}, \frac{}{100}\right.$$

$$\left.-\frac{7}{100}\right)$$

$$= \frac{1+49}{100}, \frac{3-21}{100} \ldots$$

$$= \frac{1}{2}, \frac{-9}{50}, \frac{-12}{50}, \frac{2}{5}, 0$$

ans.

(12) a_1, a_2, a_3 orthogonal basis for R^3. $\nearrow a_1 a_1 \neq 0$, $a_1 a_2 = a_1 a_3 = 0$

Show : $x_1 = \dfrac{a_1^T b}{a_1^T a_1}$

$$b = x_1 a_1 + x_2 a_2 + a_3 x_3$$

$$\Rightarrow a_1^T b = a_1^T (x_1 a_1 + x_2 a_2 + a_3 x_3)$$ $\nearrow 0$

$$= a_1^T a_1 x_1$$

$$\therefore x_1 = \frac{a_1^T b}{a_1^T a_1}$$

$= a_1 a_1$

Similar for x_2 & x_3.

(13) Given, $a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$

gotta find multiple of a to be subtracted from b to make res B orthogonal to a.

$$B = b - ka$$

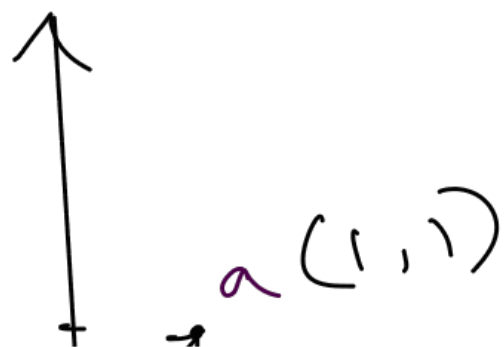
$$a^T B = 0$$

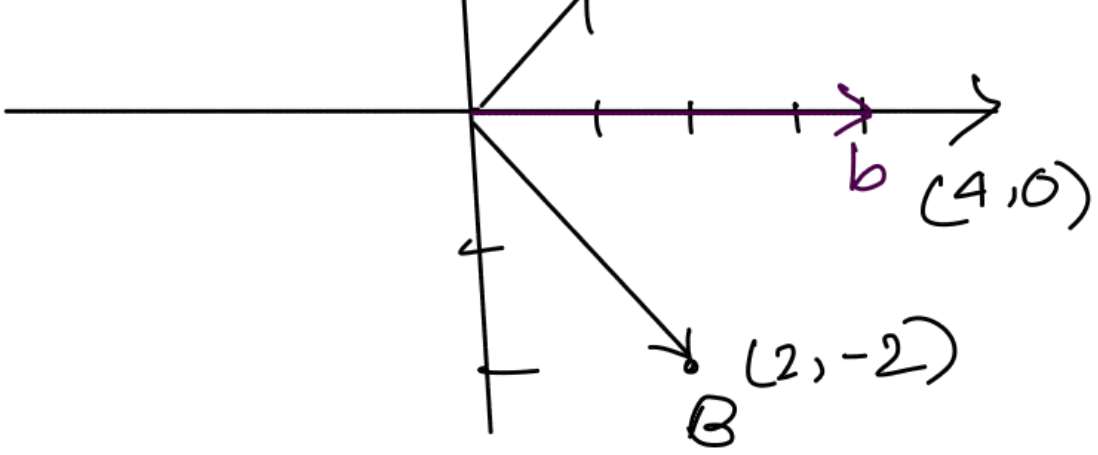
$$a^T(b - ka) = 0$$

$$a^T b = a^T a k$$

$$\therefore k = \frac{a^T b}{a^T a} = \frac{4}{2} = 2$$

$$\therefore B = \begin{bmatrix} 4 \\ 0 \end{bmatrix} - 2\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$





(4) $q_1 = \frac{a}{|a|} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix}$

S1. $q_2 = \frac{b - \text{proj}_{(q_1)} b}{|b - \text{proj}_{(q_1)} b|}$

$$\text{proj}_{(q_1)} b = (b \cdot q_1) q_1$$

$$b \cdot q_1 = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$(b \cdot q_1) q_1 = 2\sqrt{2} \begin{bmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$$

$$b - \text{proj}_{(q_1)} b = \begin{bmatrix}4\\0\end{bmatrix} - \begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}2\\-2\end{bmatrix}$$

$$\therefore q_2 = \frac{\begin{bmatrix} 2 \\ -2 \end{bmatrix}}{\sqrt{8}} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}$$

QR factorization:

$$Q = [q_1 \quad q_2]$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$R \rightarrow$ upper Triang.

$$R = \begin{bmatrix} q_1^T a & q_1^T b \\ 0 & q_2^T b \end{bmatrix}$$

$q_1^T a = \sqrt{2}$ $\quad$ $q_1^T b = 2\sqrt{2}$ $\quad$ $q_2^T b = 2\sqrt{2}$

$\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}\right)$ $\quad$ $\left(\frac{4}{\sqrt{2}}\right)$ $\quad$ same.

$$R = \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 2\sqrt{2} \end{bmatrix}$$

$$\therefore A = QR = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 2\sqrt{2} \end{bmatrix}$$

(15) $$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$$

a) O. Norm q_1 & q_2 who span C(A).

gram schimdt on cols of A! lovely.

=

(secret, A has 2 cols → Col rank 2. SO q_3 in orthogonal complement of C(A)

→ $N(A^T)$ → Left Null (Ans to b)

$$a_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

$$= \sqrt{1+4+4} = 3$$

$|a_1| = \sqrt{1+4+\quad} = 3$

$$q_1 = \frac{1}{3}\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

$$q_2 = \frac{a_2 - ka_1}{|a_2 - ka_1|}$$

$$k = \frac{a_1^T a_2}{a_1^T a_1} = \frac{-9}{9} = -1$$

$$\begin{matrix} 1+1 \\ -1+2 \\ 4-2 \end{matrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$\therefore q_2 = \frac{1}{3}\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$ Alhamdulillah!

(c) Solve $Ax = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$ by least sq.

$$A^T A x = A^T b$$

$$A^T A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -9 \\ -9 & 18 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 2 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix} = \begin{bmatrix} -9 \\ 27 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 0 & -9 \\ -9 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -9 \\ 27 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -9 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -9 \\ 18 \end{bmatrix}$$

$$-9x_2 = 18$$

$$x_2 = 2$$

$$\therefore -9x_1 = 18 - 9$$

$$\therefore x_1 = 1$$

$$\therefore x_1 x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 & 2 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 9 & -9 \\ -9 & 18 \end{bmatrix}$$

Step 2: Compute $A^T\mathbf{b}$

$$A^T\mathbf{b} = \begin{bmatrix} 1 & 2 & -2 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 1+4-14 \\ 1-2+28 \end{bmatrix} = \begin{bmatrix} -9 \\ 27 \end{bmatrix}$$

Step 3: Compute $(A^TA)^{-1}$

We already have:

$$A^TA = \begin{vmatrix} 9 & -9 \\ -9 & 18 \end{vmatrix}$$

Let's find the inverse. For a 2×2 matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Here:

- $a = 9, b = -9, c = -9, d = 18$
- Determinant = $9 \times 18 - (-9) \times (-9) = 162 - 81 = 81$

So,

$$(A^TA)^{-1} = \frac{1}{81} \begin{vmatrix} 18 & 9 \\ 9 & 9 \end{vmatrix}$$

Step 4: Now compute $\mathbf{x}_b = (A^TA)^{-1}A^T\mathbf{b}$

$$\mathbf{x}_b = \frac{1}{81} \begin{vmatrix} 18 & 9 \\ 9 & 9 \end{vmatrix} \begin{bmatrix} -9 \\ 27 \end{bmatrix} = \frac{1}{81} \begin{bmatrix} 18(-9)+ \\ 9(-9)+ \end{bmatrix} = \frac{1}{81} \begin{bmatrix} -162+243 \\ -81+243 \end{bmatrix} = \frac{1}{81} \begin{bmatrix} 81 \\ 162 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Ask anything

$x_b = (A^TA)^{-1}A^Tb$

(16) multiple of a closest to b ⇒ projection of b on a.

$$p = \frac{a^T b}{a^T a} a = \frac{14}{40}(4,5,2,2)$$

orthonormal q_1 & q_2 in plane of a, b

$$|a| = \sqrt{40} \quad \therefore \quad q_1 = \frac{a}{|a|} = \frac{1}{7}(4,5,2,2)$$

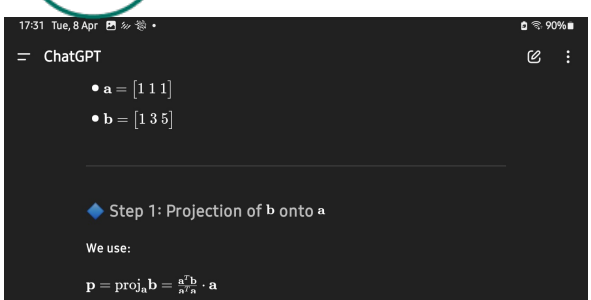
$$B = b - p = (1,2,0,0) - \frac{14}{40}(4,5,2,2)$$

$$= -\frac{1}{7}, \frac{4}{7}, -\frac{4}{7}, -\frac{4}{7}$$

$$q_2 = \frac{B}{|B|} = \left(-\frac{1}{7}, \frac{4}{7}, 1, 1\right)$$

↳ 1

(17)



Compute dot products:

- $\mathbf{a}^T\mathbf{b} = 1(1) + 1(3) + 1(5) = 9$
- $\mathbf{a}^T\mathbf{a} = 1^2 + 1^2 + 1^2 = 3$

So:

$$\mathbf{p} = \frac{9}{3} \cdot \mathbf{a} = 3 \cdot \begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}3\\3\\3\end{bmatrix}$$

✅ Projection:

$$\mathbf{p} = \begin{bmatrix}3\\3\\3\end{bmatrix}$$

🔷 Step 2: Error vector $\mathbf{e} = \mathbf{b} - \mathbf{p}$

$$\mathbf{e} = \begin{bmatrix}1\\3\\5\end{bmatrix} - \begin{bmatrix}3\\3\\3\end{bmatrix} = \begin{bmatrix}-2\\0\\2\end{bmatrix}$$

✅ Error vector:

$$\mathbf{e} = \begin{bmatrix}-2\\0\\2\end{bmatrix}$$

Ask anything

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$$\mathbf{e} = \begin{bmatrix}-2\\0\\2\end{bmatrix}$$

🔷 Step 3: Orthonormal vectors $\mathbf{g}_1$ and $\mathbf{g}_2$

$\mathbf{g}_1 = \frac{\mathbf{a}}{|\mathbf{a}|}$

$$\|\mathbf{a}\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \Rightarrow \mathbf{g}_1 = \frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\1\end{bmatrix}$$

✅ $\mathbf{g}_1 = \left|\frac{1}{\sqrt{3}}\ \frac{1}{\sqrt{3}}\ \frac{1}{\sqrt{3}}\right|$

$\mathbf{g}_2 = \frac{\mathbf{e}}{|\mathbf{e}|}$

$$\|\mathbf{e}\| = \sqrt{(-2)^2 + 0^2 + 2^2} = \sqrt{8} = 2\sqrt{2} \Rightarrow \mathbf{g}_2 = \frac{1}{2\sqrt{2}}\begin{bmatrix}-2\\0\\2\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}-1\\0\\1\end{bmatrix}$$

✅ $\mathbf{g}_2 = \left|\frac{-1}{\sqrt{2}}\ 0\ \frac{1}{\sqrt{2}}\right|$

✅ Final Answers:

- Projection $\mathbf{p} = [3\ 3\ 3]$
- Error vector $\mathbf{e} = [-2\ 0\ 2]$
- Orthonormal basis vectors:
  - $\mathbf{g}_1 = \frac{1}{\sqrt{3}}[1\ 1\ 1]$
  - $\mathbf{g}_2 = \frac{1}{\sqrt{2}}[-1\ 0\ 1]$

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Let:

- $\mathbf{a} = (1, -1, 0, 0)$
- $\mathbf{b} = (0, 1, -1, 0)$
- $\mathbf{c} = (0, 0, 1, -1)$

We will apply Gram-Schmidt to obtain orthogonal vectors A, B, C that span the same space and are perpendicular to $\mathbf{d} = (1, 1, 1, 1)$.

🔷 Step 1: Let $\mathbf{A} = \mathbf{a}$

Nothing to change here — the first vector in the orthogonal basis is simply:

$$\begin{bmatrix}1\\1\end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

🔹 Step 2: Make $\mathbf{B}$ orthogonal to $\mathbf{A}$

$$\mathbf{B} = \mathbf{b} - \text{proj}_{\mathbf{A}}\mathbf{b}$$

Compute:

- $\mathbf{A}^T\mathbf{b} = 1 \cdot 0 + (-1) \cdot 1 = -1$
- $\mathbf{A}^T\mathbf{A} = 1^2 + (-1)^2 = 2$

$$\text{proj}_{\mathbf{A}}\mathbf{b} = \frac{-1}{2} \cdot \mathbf{A} = \frac{-1}{2} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 0 \end{bmatrix}$$

So:

$$\mathbf{B} = \mathbf{b} - \text{proj}_{\mathbf{A}}\mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \\ 0 \end{bmatrix}$$

✅ So:

$$\mathbf{B} = \left(\tfrac{1}{2}, \tfrac{1}{2}, -1, 0\right)$$

🔹 Step 3: Make $\mathbf{C}$ orthogonal to both $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{C} = \mathbf{c} - \text{proj}_{\mathbf{A}}\mathbf{c} - \text{proj}_{\mathbf{B}}\mathbf{c}$$

First projection:

- $\mathbf{A}^T\mathbf{c} = 1 \cdot 0 + (-1) \cdot 0 = 0 \Rightarrow \text{proj}_{\mathbf{A}}\mathbf{c} = 0$

Second projection:

- $\mathbf{B}^T\mathbf{c} = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0 + (-1) \cdot 1 = -1$
- $\mathbf{B}^T\mathbf{B} = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + 1^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$

$$\text{proj}_{\mathbf{B}}\mathbf{c} = \frac{-1}{3/2} \cdot \mathbf{B} = -\frac{2}{3} \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$$

So:

$$\mathbf{C} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ -1 \end{bmatrix}$$

✅ So:

$$\mathbf{C} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, -1\right)$$

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$$\mathbf{B} = \mathbf{b} - \text{proj}_{\mathbf{A}}\mathbf{b} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \\ 0 \end{bmatrix}$$

✅ So:

$$\mathbf{B} = \left(\tfrac{1}{2}, \tfrac{1}{2}, -1, 0\right)$$

🔹 Step 3: Make $\mathbf{C}$ orthogonal to both $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{C} = \mathbf{c} - \text{proj}_{\mathbf{A}}\mathbf{c} - \text{proj}_{\mathbf{B}}\mathbf{c}$$

First projection:

- $\mathbf{A}^T\mathbf{c} = 1 \cdot 0 + (-1) \cdot 0 = 0 \Rightarrow \text{proj}_{\mathbf{A}}\mathbf{c} = 0$

Second projection:

- $\mathbf{B}^T\mathbf{c} = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0 + (-1) \cdot 1 = -1$
- $\mathbf{B}^T\mathbf{B} = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + 1^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$

$$\text{proj}_{\mathbf{B}}\mathbf{c} = \frac{-1}{3/2} \cdot \mathbf{B} = -\frac{2}{3} \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$$

So:

$$\mathbf{C} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ -1 \end{bmatrix}$$

✅ So:

$$\mathbf{C} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, -1\right)$$

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Second projection:

- $\mathbf{B}^T\mathbf{c} = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 0 + (-1) \cdot 1 = -1$
- $\mathbf{B}^T\mathbf{B} = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + 1^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$

$$\text{proj}_{\mathbf{B}}\mathbf{c} = \frac{-1}{3/2} \cdot \mathbf{B} = -\frac{2}{3} \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$$

So:

$$\mathbf{C} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \\ -1 \end{bmatrix}$$

✅ So:

$$\mathbf{C} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, -1\right)$$

✅ Final Orthogonal Vectors:

- $\mathbf{A} = (1, -1, 0, 0)$
- $\mathbf{B} = \left(\frac{1}{2}, \frac{1}{2}, -1, 0\right)$
- $\mathbf{C} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1\right)$

✳️ Observation:

- All three are orthogonal to $\mathbf{d} = (1, 1, 1, 1)$
- They span the 3D hyperplane in $\mathbb{R}^4$ perpendicular to $\mathbf{d}$

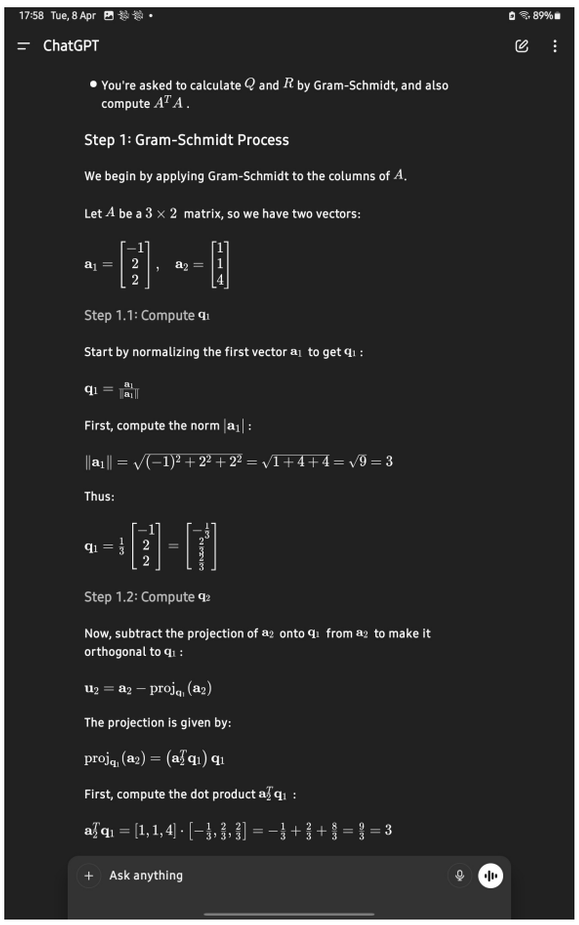
Want to continue and normalize them into an orthonormal set $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$?

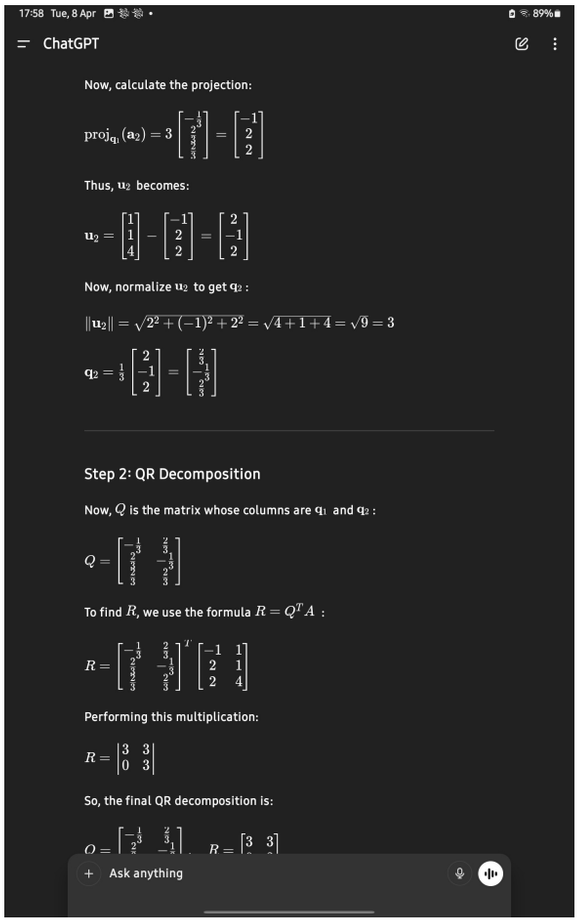
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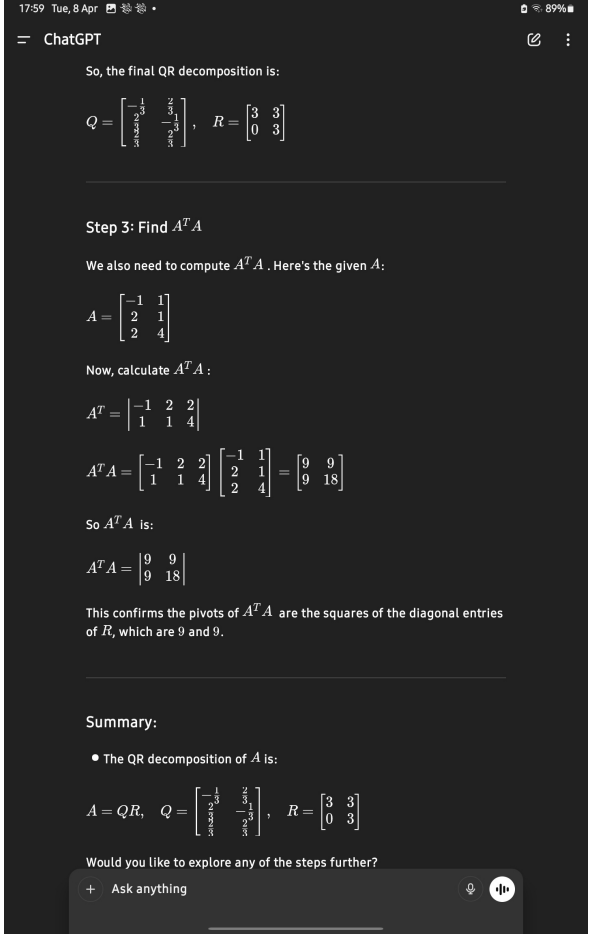
(19) $A = QR$ ∴ $A^TA = R^TQ^TQR$

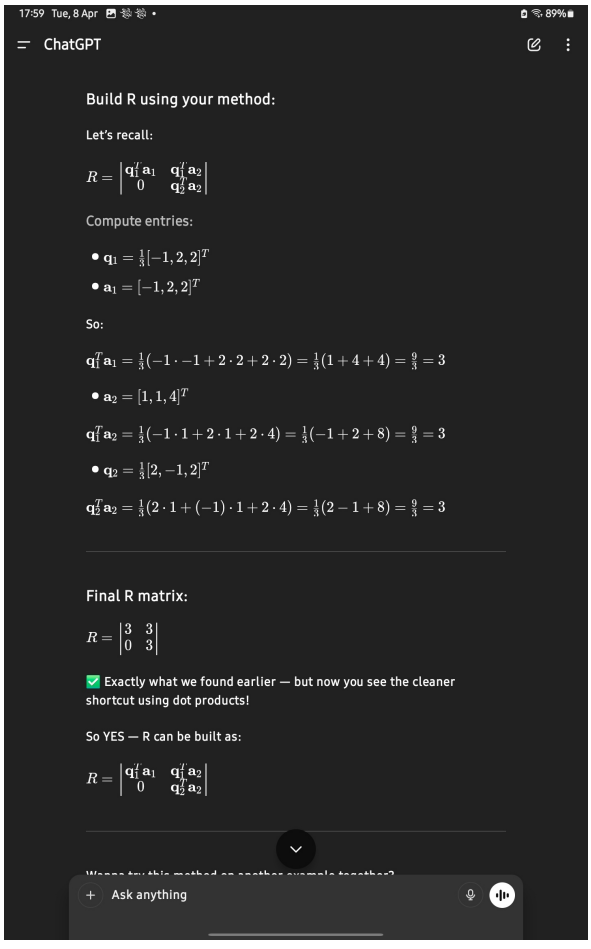
$= R^TR$

= lower trig times upper trigngle.



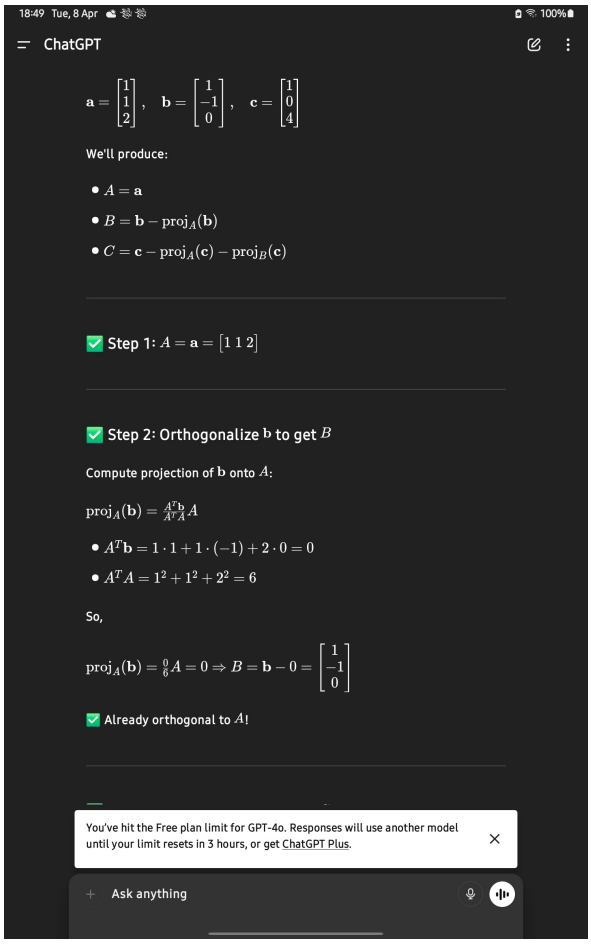


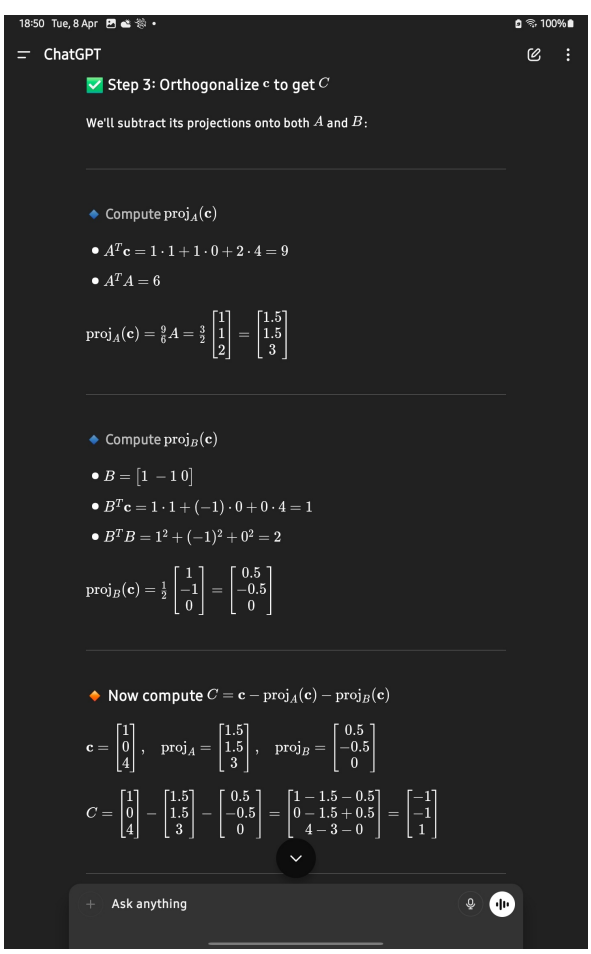

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So, the final QR decomposition is:
$Q = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{2}{3} \end{bmatrix}, \quad R = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$
Step 3: Find A^TA
We also need to compute A^TA. Here's the given A:
$A = \begin{bmatrix} -1 & 1 \\ 2 & 1 \\ 2 & 4 \end{bmatrix}$
Now, calculate A^TA:
$A^T = \begin{vmatrix} -1 & 2 & 2 \\ 1 & 1 & 4 \end{vmatrix}$
$A^TA = \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 9 & 9 \\ 9 & 18 \end{bmatrix}$
So A^TA is:
$A^TA = \begin{vmatrix} 9 & 9 \\ 9 & 18 \end{vmatrix}$
This confirms the pivots of A^TA are the squares of the diagonal entries of R, which are 9 and 9.
Summary:
• The QR decomposition of A is:
$A = QR, \quad Q = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{2}{3} \end{bmatrix}, \quad R = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$
Would you like to explore any of the steps further?
Ask anything


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Build R using your method:
Let's recall:
$R = \begin{vmatrix} \mathbf{q}_1^T\mathbf{a}_1 & \mathbf{q}_1^T\mathbf{a}_2 \\ 0 & \mathbf{q}_2^T\mathbf{a}_2 \end{vmatrix}$
Compute entries:
• $\mathbf{q}_1 = \frac{1}{3}[-1, 2, 2]^T$
• $\mathbf{a}_1 = [-1, 2, 2]^T$
So:
$\mathbf{q}_1^T\mathbf{a}_1 = \frac{1}{3}(-1 \cdot -1 + 2 \cdot 2 + 2 \cdot 2) = \frac{1}{3}(1 + 4 + 4) = \frac{9}{3} = 3$
• $\mathbf{a}_2 = [1, 1, 4]^T$
$\mathbf{q}_1^T\mathbf{a}_2 = \frac{1}{3}(-1 \cdot 1 + 2 \cdot 1 + 2 \cdot 4) = \frac{1}{3}(-1 + 2 + 8) = \frac{9}{3} = 3$
• $\mathbf{q}_2 = \frac{1}{3}[2, -1, 2]^T$
$\mathbf{q}_2^T\mathbf{a}_2 = \frac{1}{3}(2 \cdot 1 + (-1) \cdot 1 + 2 \cdot 4) = \frac{1}{3}(2 - 1 + 8) = \frac{9}{3} = 3$
Final R matrix:
$R = \begin{vmatrix} 3 & 3 \\ 0 & 3 \end{vmatrix}$
✅ Exactly what we found earlier — but now you see the cleaner shortcut using dot products!
So YES — R can be built as:
$R = \begin{vmatrix} \mathbf{q}_1^T\mathbf{a}_1 & \mathbf{q}_1^T\mathbf{a}_2 \\ 0 & \mathbf{q}_2^T\mathbf{a}_2 \end{vmatrix}$
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22.


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$\mathbf{a} = \begin{bmatrix}1\\1\\2\end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix}1\\-1\\0\end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix}1\\0\\4\end{bmatrix}$
We'll produce:
• $A = \mathbf{a}$
• $B = \mathbf{b} - \text{proj}_A(\mathbf{b})$
• $C = \mathbf{c} - \text{proj}_A(\mathbf{c}) - \text{proj}_B(\mathbf{c})$
✅ Step 1: $A = \mathbf{a} = [1\ 1\ 2]$
✅ Step 2: Orthogonalize $\mathbf{b}$ to get B
Compute projection of $\mathbf{b}$ onto A:
$\text{proj}_A(\mathbf{b}) = \frac{A^T\mathbf{b}}{A^TA}A$
• $A^T\mathbf{b} = 1 \cdot 1 + 1 \cdot (-1) + 2 \cdot 0 = 0$
• $A^TA = 1^2 + 1^2 + 2^2 = 6$
So,
$\text{proj}_A(\mathbf{b}) = \frac{0}{6}A = 0 \Rightarrow B = \mathbf{b} - 0 = \begin{bmatrix}1\\-1\\0\end{bmatrix}$
✅ Already orthogonal to A!
You've hit the Free plan limit for GPT-4o. Responses will use another model until your limit resets in 3 hours, or get ChatGPT Plus.
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✅ Step 3: Orthogonalize $\mathbf{c}$ to get C
We'll subtract its projections onto both A and B:
🔹 Compute $\text{proj}_A(\mathbf{c})$
• $A^T\mathbf{c} = 1 \cdot 1 + 1 \cdot 0 + 2 \cdot 4 = 9$
• $A^TA = 6$
$\text{proj}_A(\mathbf{c}) = \frac{9}{6}A = \frac{3}{2}\begin{bmatrix}1\\1\\2\end{bmatrix} = \begin{bmatrix}1.5\\1.5\\3\end{bmatrix}$
🔹 Compute $\text{proj}_B(\mathbf{c})$
• $B = [1\ -1\ 0]$
• $B^T\mathbf{c} = 1 \cdot 1 + (-1) \cdot 0 + 0 \cdot 4 = 1$
• $B^TB = 1^2 + (-1)^2 + 0^2 = 2$
$\text{proj}_B(\mathbf{c}) = \frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}0.5\\-0.5\\0\end{bmatrix}$
🔸 Now compute $C = \mathbf{c} - \text{proj}_A(\mathbf{c}) - \text{proj}_B(\mathbf{c})$
$\mathbf{c} = \begin{bmatrix}1\\0\\4\end{bmatrix}, \quad \text{proj}_A = \begin{bmatrix}1.5\\1.5\\3\end{bmatrix}, \quad \text{proj}_B = \begin{bmatrix}0.5\\-0.5\\0\end{bmatrix}$
$C = \begin{bmatrix}1\\0\\4\end{bmatrix} - \begin{bmatrix}1.5\\1.5\\3\end{bmatrix} - \begin{bmatrix}0.5\\-0.5\\0\end{bmatrix} = \begin{bmatrix}1-1.5-0.5\\0-1.5+0.5\\4-3-0\end{bmatrix} = \begin{bmatrix}-1\\-1\\1\end{bmatrix}$
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Given Matrix A:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$$

Step 1: Apply the Gram-Schmidt Process

Let the columns of A be:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Step 1.1: Find q_1

Start by setting $q_1 = \mathbf{a}_1$ and normalizing it:

$$\|\mathbf{a}_1\| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

$$q_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Step 1.2: Find q_2

Now we subtract the projection of $\mathbf{a}_2$ onto q_1 from $\mathbf{a}_2$:

$$\text{proj}_{q_1}(\mathbf{a}_2) = \frac{q_1^T \mathbf{a}_2}{q_1^T q_1} q_1$$

- $q_1^T \mathbf{a}_2 = 1 \cdot 2 + 0 \cdot 0 + 0 \cdot 3 = 2$
- $q_1^T q_1 = 1^2 + 0^2 + 0^2 = 1$

$$\text{proj}_{q_1}(\mathbf{a}_2) = 2 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

Now, subtract this projection from $\mathbf{a}_2$:

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Now, subtract this projection from $\mathbf{a}_2$:

$$\mathbf{a}_2' = \mathbf{a}_2 - \text{proj}_{q_1}(\mathbf{a}_2) = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

Now normalize $\mathbf{a}_2'$:

$$\|\mathbf{a}_2'\| = \sqrt{0^2 + 0^2 + 3^2} = 3$$

$$q_2 = \frac{\mathbf{a}_2'}{\|\mathbf{a}_2'\|} = \frac{1}{3} \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Step 1.3: Find q_3

Now we subtract the projections of $\mathbf{a}_3$ onto both q_1 and q_2:

$$\text{proj}_{q_1}(\mathbf{a}_3) = \frac{q_1^T \mathbf{a}_3}{q_1^T q_1} q_1 = \frac{1 \cdot 4 + 0 \cdot 5 + 0 \cdot 6}{1} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{proj}_{q_2}(\mathbf{a}_3) = \frac{q_2^T \mathbf{a}_3}{q_2^T q_2} q_2 = \frac{0 \cdot 4 + 0 \cdot 5 + 1 \cdot 6}{1} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}$$

Now subtract these projections from $\mathbf{a}_3$:

$$\mathbf{a}_3' = \mathbf{a}_3 - \text{proj}_{q_1}(\mathbf{a}_3) - \text{proj}_{q_2}(\mathbf{a}_3) = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$$

Now normalize $\mathbf{a}_3'$:

$$\|\mathbf{a}_3'\| = \sqrt{0^2 + 5^2 + 0^2} = 5$$

$$q_3 = \frac{\mathbf{a}_3'}{\|\mathbf{a}_3'\|} = \frac{1}{5} \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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Now normalize $\mathbf{a}_3'$:

$$\|\mathbf{a}_3'\| = \sqrt{0^2 + 5^2 + 0^2} = 5$$

$$q_3 = \frac{\mathbf{a}_3'}{\|\mathbf{a}_3'\|} = \frac{1}{5} \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Step 2: Construct Q and R

We now have the orthonormal vectors:

$$q_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad q_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad q_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Now, construct the matrix Q:

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Next, we compute R. Since $A = QR$, we find:

$$R = Q^T A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 5 \\ 0 & 3 & 6 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

Final Answer:

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

Thus, $A = QR$.

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(a) Find a basis for the subspace S in $\mathbb{R}^4$ spanned by all solutions of the equation:

$$x_1 + x_2 + x_3 - x_4 = 0$$

We want to find the solutions of the equation:

$$x_1 + x_2 + x_3 - x_4 = 0$$

This is a linear equation with four variables, so we can express some of the variables in terms of others. Let's solve for x_1 in terms of x_2, x_3, and x_4:

$$x_1 = -x_2 - x_3 + x_4$$

Thus, the general solution is:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -x_2 - x_3 + x_4 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

This can be written as:

$$\mathbf{x} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Therefore, a basis for the subspace S is:

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(b) Find a basis for the orthogonal complement of S, denoted $S^\perp$.

The subspace $S^\perp$ consists of all vectors in $\mathbb{R}^4$ that are orthogonal to every vector in S. To find a basis for $S^\perp$, we need to find the set of vectors that are orthogonal to all three basis vectors of S.

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(b) Find a basis for the orthogonal complement of S, denoted $S^\perp$.

The subspace $S^\perp$ consists of all vectors in $\mathbb{R}^4$ that are orthogonal to every vector in S. To find a basis for $S^\perp$, we need to find the set of vectors that are orthogonal to all three basis vectors of S.

Let the vector $\mathbf{y} = \begin{bmatrix} y_1 & y_2 & y_3 & y_4 \end{bmatrix}$ belong to $S^\perp$. For $\mathbf{y}$ to be orthogonal to all three vectors in S, we must have the following dot products equal to 0:

$$\mathbf{y}^T \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = -y_1 + y_2 = 0$$

$$\mathbf{y}^T \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = -y_1 + y_3 = 0$$

$$\mathbf{y}^T \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = y_1 + y_4 = 0$$

From the equations above, we get:

1. $-y_1 + y_2 = 0 \quad \Rightarrow \quad y_2 = y_1$
2. $-y_1 + y_3 = 0 \quad \Rightarrow \quad y_3 = y_1$
3. $y_1 + y_4 = 0 \quad \Rightarrow \quad y_4 = -y_1$

Thus, $\mathbf{y} = \begin{bmatrix} y_1 & y_1 & y_1 & -y_1 \end{bmatrix} = y_1 \begin{bmatrix} 1 & 1 & 1 & -1 \end{bmatrix}$.

Therefore, a basis for $S^{\perp}$ is:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

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(c) Find $\mathbf{b}_1$ in S and $\mathbf{b}_2$ in $S^{\perp}$ such that $\mathbf{b}_1 + \mathbf{b}_2 = \mathbf{b} = (1, 1, 1, 1)$.

Step 1: Find $\mathbf{b}_2$, the component of $\mathbf{b}$ in $S^{\perp}$.

To find $\mathbf{b}_2$, we compute the projection of $\mathbf{b} = (1, 1, 1, 1)$ onto the subspace $S^{\perp}$. The projection formula is:

$$\mathbf{b}_2 = \text{proj}_{S^{\perp}}(\mathbf{b}) = \frac{\mathbf{b}^T \mathbf{v}}{\mathbf{v}^T \mathbf{v}} \mathbf{v}$$

where $\mathbf{v} = \begin{bmatrix} 1 & 1 & 1 & -1 \end{bmatrix}$ is the basis vector of $S^{\perp}$.

We compute the dot products:

$$\mathbf{b}^T \mathbf{v} = 1(1) + 1(1) + 1(1) + 1(-1) = 2$$

\mathbf{v}^T \mathbf{v} = 1(1) + 1(1) + 1(1) + (-1)(-1) = 4]

Thus,

$$\mathbf{b}_2 = \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

So,

$$\mathbf{b}_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

Step 2: Find $\mathbf{b}_1$.

Now we find $\mathbf{b}_1$ as:

$$\mathbf{b}_1 = \mathbf{b} - \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$\mathbf{b}_1 = \begin{bmatrix} 1 - \frac{1}{2} \\ 1 - \frac{1}{2} \\ 1 - \frac{1}{2} \\ 1 + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$$

Ask anything

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We compute the dot products:

$$\mathbf{b}^T \mathbf{v} = 1(1) + 1(1) + 1(1) + 1(-1) = 2$$

\mathbf{v}^T \mathbf{v} = 1(1) + 1(1) + 1(1) + (-1)(-1) = 4]

Thus,

$$\mathbf{b}_2 = \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

So,

$$\mathbf{b}_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

Step 2: Find $\mathbf{b}_1$.

Now we find $\mathbf{b}_1$ as:

$$\mathbf{b}_1 = \mathbf{b} - \mathbf{b}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$\mathbf{b}_1 = \begin{bmatrix} 1 - \frac{1}{2} \\ 1 - \frac{1}{2} \\ 1 - \frac{1}{2} \\ 1 + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$$

Check:

Finally, let's check if $\mathbf{b}_1 + \mathbf{b}_2 = \mathbf{b}$:

$$\mathbf{b}_1 + \mathbf{b}_2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Thus, the components $\mathbf{b}_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{3}{2} \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$ satisfy $\mathbf{b}_1 + \mathbf{b}_2 = \mathbf{b}$.

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Given Matrix A:

1. Case 1:

$$A = \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}$$

Gram-Schmidt Process:

1. Step 1: Find q_1 : We start with the first column of A, which is $\mathbf{a}_1 = [2\ 1]$.

   The norm of $\mathbf{a}_1$ is:

$$\|\mathbf{a}_1\| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$q_1 = \frac{1}{\|\mathbf{a}_1\|}\mathbf{a}_1 = \frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{vmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{vmatrix}$$

1. Step 2: Find q_2 : The second column of A is $\mathbf{a}_2 = [1\ 1]$. We first subtract the projection of $\mathbf{a}_2$ onto q_1 .

   The projection of $\mathbf{a}_2$ onto q_1 is:

$$\text{proj}_{q_1}(\mathbf{a}_2) = (q_1^T \mathbf{a}_2)q_1 = \left(\frac{2}{\sqrt{5}} \cdot 1 + \frac{1}{\sqrt{5}} \cdot 1\right) \begin{vmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{vmatrix} = \frac{3}{\sqrt{5}} \begin{vmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{vmatrix} = \frac{3}{5}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$v_2 = \mathbf{a}_2 - \text{proj}_{q_1}(\mathbf{a}_2) = \begin{vmatrix} 1 \\ 1 \end{vmatrix} - \frac{3}{5}\begin{vmatrix} 2 \\ 1 \end{vmatrix} = \begin{vmatrix} 1 - \frac{6}{5} \\ 1 - \frac{3}{5} \end{vmatrix} = \begin{vmatrix} -\frac{1}{5} \\ \frac{2}{5} \end{vmatrix}$$

$$\|v_2\| = \sqrt{\left(-\frac{1}{5}\right)^2 + \left(\frac{2}{5}\right)^2} = \sqrt{\frac{1}{25} + \frac{4}{25}} = \sqrt{\frac{5}{25}} = \frac{\sqrt{5}}{5}$$

$$q_2 = \frac{1}{\|v_2\|}v_2 = \frac{5}{\sqrt{5}} \begin{vmatrix} -\frac{1}{5} \\ \frac{2}{5} \end{vmatrix} = \frac{1}{\sqrt{5}} \begin{vmatrix} -1 \\ 2 \end{vmatrix}$$

1. Step 3: Form the matrix Q and R:

   Now, we have:

$$Q = \begin{vmatrix} \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{vmatrix}$$

$$R = \begin{bmatrix} q_1^T \mathbf{a}_1 & q_1^T \mathbf{a}_2 \\ 0 & q_2^T \mathbf{a}_2 \end{bmatrix} = \begin{vmatrix} \sqrt{5} & \frac{3}{\sqrt{5}} \\ 0 & \frac{\sqrt{5}}{5} \end{vmatrix}$$

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$$\|v_2\| = \sqrt{\left(-\frac{1}{5}\right)^2 + \left(\frac{2}{5}\right)^2} = \sqrt{\frac{1}{25} + \frac{4}{25}} = \sqrt{\frac{5}{25}} = \frac{\sqrt{5}}{5}$$

$$q_2 = \frac{1}{\|v_2\|}v_2 = \frac{5}{\sqrt{5}} \begin{vmatrix} -\frac{1}{5} \\ \frac{2}{5} \end{vmatrix} = \frac{1}{\sqrt{5}} \begin{vmatrix} -1 \\ 2 \end{vmatrix}$$

1. Step 3: Form the matrix Q and R:

   Now, we have:

$$Q = \begin{vmatrix} \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{vmatrix}$$

$$R = \begin{bmatrix} q_1^T \mathbf{a}_1 & q_1^T \mathbf{a}_2 \\ 0 & q_2^T \mathbf{a}_2 \end{bmatrix} = \begin{vmatrix} \sqrt{5} & \frac{3}{\sqrt{5}} \\ 0 & \frac{\sqrt{5}}{5} \end{vmatrix}$$

So the QR decomposition for A is:

$$A = QR = \begin{vmatrix} \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{vmatrix} \begin{vmatrix} \sqrt{5} & \frac{3}{\sqrt{5}} \\ 0 & \frac{\sqrt{5}}{5} \end{vmatrix}$$

Case 2: $A = \begin{bmatrix} 1 & 1\,1 & 1\,1 & 1 \end{bmatrix}$

For the case when the columns are dependent (i.e., all columns are equal), the Gram-Schmidt process will break down because the second column will become a multiple of the first, and thus R_{22} (the entry in the second row, second column of R) will become zero.

This happens because:

$$R = \begin{vmatrix} q_1^T \mathbf{a}_1 & q_1^T \mathbf{a}_2 \\ 0 & q_2^T \mathbf{a}_2 \end{vmatrix}$$

Conclusion:

- In the case where the columns of A are dependent, R_{22} becomes zero, and the Gram-Schmidt process breaks down because you cannot normalize the second vector.

Ask anything

26 $(q_1^T c^*)\, q_2 = \dfrac{B^T c}{\quad} B$

B^TB

c2 $q_2 = \frac{B}{|B|}$ & extra q_1 in C^*

is orthogo to q_2

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We are given the matrix:

$$Q = c\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}$$

We want to make sure the columns of Q are orthonormal (i.e., each column should have a magnitude of 1, and the columns should be orthogonal to each other).

To achieve orthonormal columns, we need to normalize each column vector. If you look at the matrix, each column vector has the same entries but with different signs.

To normalize the first column:

The first column is $(1, -1, -1, -1)^\top$, and we compute its magnitude:

$$\|(1, -1, -1, -1)\| = \sqrt{1^2 + (-1)^2 + (-1)^2 + (-1)^2} = \sqrt{4} = 2$$

So, to normalize this column, we divide each element by 2:

$$q_1 = \frac{1}{2}\begin{bmatrix}1\\-1\\-1\\-1\end{bmatrix}$$

We can do this for each column of the matrix. Since the structure of the matrix is symmetric and each column is of the same magnitude, choosing $c = \frac{1}{2}$ will ensure that all columns are normalized (i.e., each column will have a magnitude of 1).

Thus, the matrix becomes:

$$Q = \frac{1}{2}\begin{bmatrix}1 & -1 & -1 & -1\\-1 & 1 & -1 & -1\\-1 & -1 & 1 & -1\\-1 & -1 & -1 & 1\end{bmatrix}$$

Step 2: Project $\mathbf{b}$ onto the first column and then the plane of the first two columns

Next, you are asked to project $\mathbf{b} = (1,1,1,1)^\top$ onto the first column of Q, and then onto the plane spanned by the first two columns.

Ask anything

We already know the first column of Q is:

$$q_1 = \frac{1}{2}\begin{bmatrix}1\\-1\\-1\\-1\end{bmatrix}$$

The formula for the projection of $\mathbf{b}$ onto q_1 is:

$$\text{proj}_{q_1}\mathbf{b} = (q_1^\top \mathbf{b})q_1$$

Now compute $q_1^\top \mathbf{b}$:

$$q_1^\top \mathbf{b} = \tfrac{1}{2}(1\cdot 1 + (-1)\cdot 1 + (-1)\cdot 1 + (-1)\cdot 1) = \tfrac{1}{2}(1-1-1-1) = \tfrac{-2}{2} = -1$$

Thus, the projection is:

$$\text{proj}_{q_1}\mathbf{b} = -q_1 = -\frac{1}{2}\begin{bmatrix}1\\-1\\-1\\-1\end{bmatrix} = \begin{bmatrix}-\frac{1}{2}\\\frac{1}{2}\\\frac{1}{2}\\\frac{1}{2}\end{bmatrix}$$

Projection onto the plane of the first two columns:

Next, we need to project $\mathbf{b}$ onto the plane spanned by the first two columns of Q. This involves projecting onto the span of both q_1 and q_2, which are the first two orthonormal columns of Q.

The second column of Q is:

$$q_2 = \frac{1}{2}\begin{bmatrix}-1\\1\\-1\\-1\end{bmatrix}$$

To project $\mathbf{b}$ onto both q_1 and q_2, you compute the projections onto q_1 and q_2, and then add them:

$$\text{proj}_{q_1}\mathbf{b} = -q_1, \quad \text{proj}_{q_2}\mathbf{b} = -q_2$$

Thus, the final projection onto the plane is:

$$\text{proj}_{\text{plane}}\mathbf{b} = -q_1 - q_2 = \begin{bmatrix}-\frac{1}{2}\\\frac{1}{2}\\\frac{1}{2}\\\frac{1}{2}\end{bmatrix} + \begin{bmatrix}\frac{1}{2}\\-\frac{1}{2}\\\frac{1}{2}\\\frac{1}{2}\end{bmatrix} = \begin{bmatrix}0\\0\\1\\1\end{bmatrix}$$

Ask anything

And we are tasked with:

1. Choose c so that Q is orthogonal.
2. Project $\mathbf{b} = (1,1,1,1)$ onto the first column, then the plane of the first two columns.

Step 1: Making Q orthogonal

To make Q orthogonal, its columns must be orthonormal.

We choose $c = \frac{1}{2}$ to ensure that the columns are orthonormal, based on the result you mentioned earlier.

So,

$$Q = \frac{1}{2}\begin{bmatrix}1 & -1 & -1 & -1\\-1 & 1 & -1 & -1\\-1 & -1 & 1 & -1\\-1 & -1 & -1 & 1\end{bmatrix}$$

Now, the columns of Q are orthonormal.

Step 2: Project $\mathbf{b} = (1,1,1,1)$ onto the first column of Q

Let the first column of Q be:

$$q_1 = \tfrac{1}{2}(1,-1,-1,-1)$$

Now, let's compute the projection of $\mathbf{b} = (1,1,1,1)$ onto q_1 using the formula:

$$\text{proj}_{q_1}\mathbf{b} = \frac{q_1^\top b}{q_1^\top q_1}q_1$$

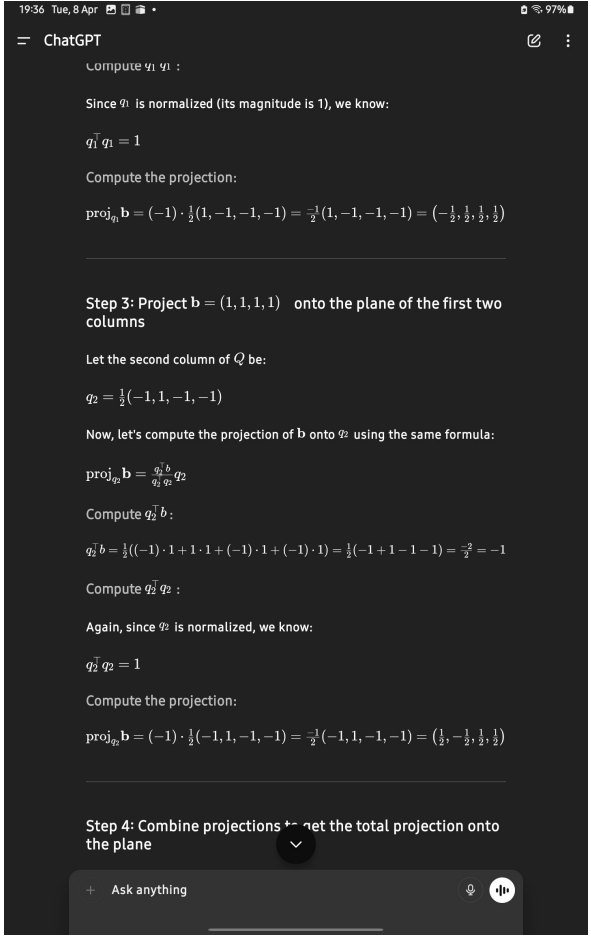
Compute $q_1^\top b$:

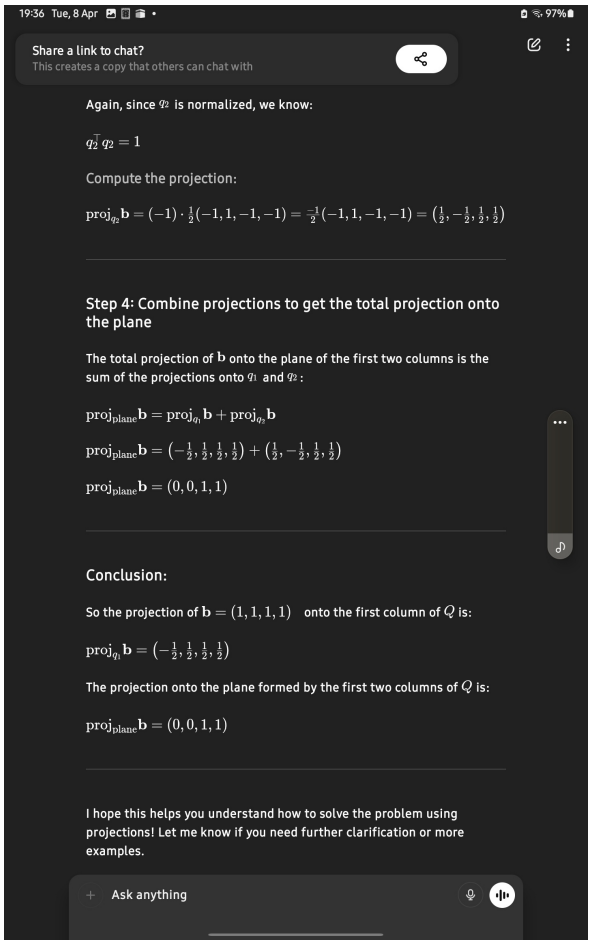
$$q_1^\top b = \tfrac{1}{2}(1\cdot 1 + (-1)\cdot 1 + (-1)\cdot 1 + (-1)\cdot 1) = \tfrac{1}{2}(1-1-1-1) = \tfrac{-2}{2} = -1$$

Compute $q_1^\top q_1$:

Since q_1 is normalized (its magnitude is 1), we know:

Ask anything

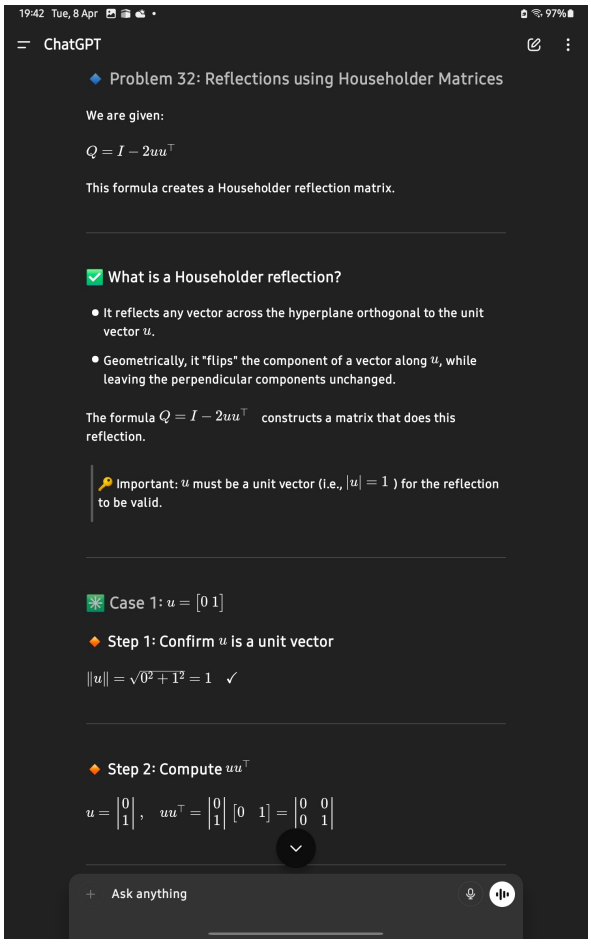


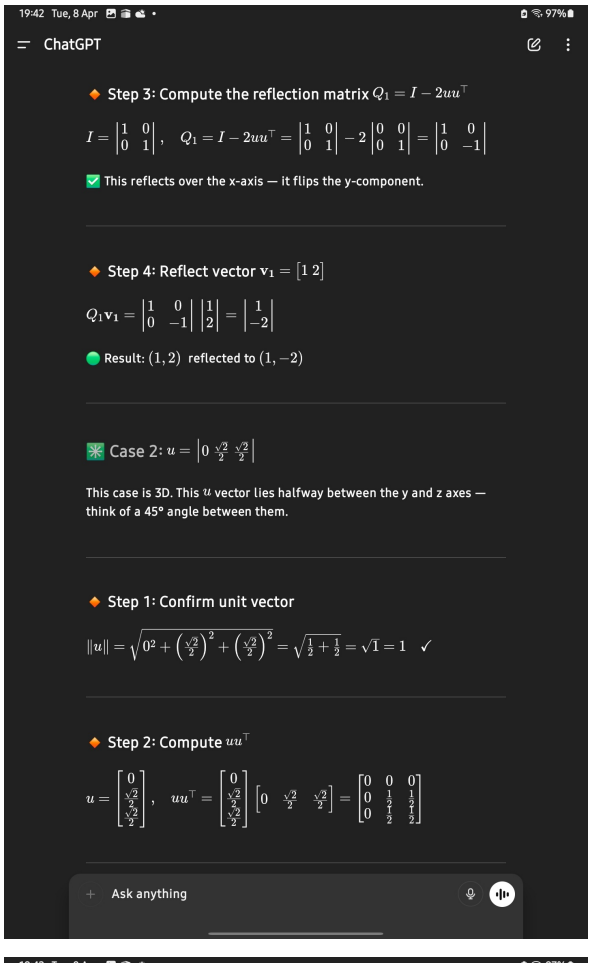


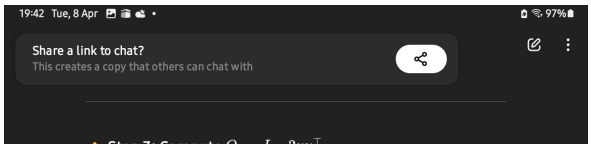
32

$Q - T - 2uu^T$ → Reflection using

Householder Matrices







◆ Step 3: Compute $Q_2 = I - 2uu^T$

$$Q_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

✅ This reflects across the plane $x_2 = x_3$ — it swaps and flips y and z axes.

◆ Step 4: Reflect vector $\mathbf{v_2} = [1\ 1\ 1]$

$$Q_2\mathbf{v_2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

🟢 Result: $(1, 1, 1)$ reflected to $(1, -1, -1)$

⬅ Final Summary of Problem 32

Case	Unit Vector u	Reflection Matrix Q	Effect	Result on Given Vector
1	$[0\ 1]$	$[1\ \ 0\ 0\ \ -1]$	Reflect across x-axis	$(1,2) \to (1,-2)$
2	$\left\|0\ \frac{\sqrt{2}}{2}\ \frac{\sqrt{2}}{2}\right\|$	[illegible]	Reflect across plane $x_2 = x_3$	$(1,1,1) \to (1,-1,-1)$

Ask anything

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(33) $\begin{bmatrix} +1 & 0 \\ 0 & +1 \end{bmatrix}$ ⋱

(34) $Q = I - 2uu^T$, $u^Tu = 1$, $Q^2 = I$

a) $Qu = (I - 2uu^T)u$

$= Iu - 2u(u^Tu)$ (with $u^Tu = 1$)

$= u - 2u = -u$ ✓

Reflecting a stick into a mirror (flip itself)

b) $Qv = (I - 2uu^T)v$

→ 0

$$= Iv - 2u(u^T v)$$

$$= v$$

$$Q_v = v$$

v is on the mirror so reflting doesn't move.